Solving Equations: Distributive Property & Variables on Both Sides

Easy level · 30 problems · A worked example comes before each type. Show every step.

Name: _____ Date: _____

Key Idea

Goal: get x alone, keeping the equation balanced.

- **Parentheses?** Distribute first: $a(x+b) = ax + ab$.
- x **on both sides?** Subtract one x-term so that x appears on only one side.
- **Then** undo $+/-$ first, and $\times/\div$ last. **Always** do the same thing to both sides, and check your answer.

Common Mistake

Distribute to *every* term inside the parentheses: $3(x+4) = 3x + 12$, **not** $3x + 4$.
When moving an x-term, subtract it from *both* sides: from $5x = 2x + 9$, $3x = 9$ (not $7x = 9$).

Type 1 Distribute First

Watch Me — Worked Example

Solve $3(x+2) = 21$**.**

Step 1 Distribute the 3 to *both* terms	$3x + 6 = 21$
Step 2 Undo the $+6$ (subtract 6 from both sides)	$3x = 15$
Step 3 Undo the $\times 3$ (divide both sides by 3)	$\boldsymbol{x = 5}$
Check	$3(5+2) = 3 \cdot 7 = 21$ ✓

1. $2(x+3) = 14$

$x =$ _____

2. $3(x-2) = 12$

$x =$ _____

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3. $4(x+1)=20$

$x=$ _____

4. $5(x-3)=10$

$x=$ _____

5. $2(x+5)=18$

$x=$ _____

6. $3(x+4)=27$

$x=$ _____

7. $6(x-1)=24$

$x=$ _____

8. $4(x-5)=8$

$x=$ _____

Type 2 Variables on Both Sides

Watch Me — Worked Example

Solve $6x+4=2x+20$**.**

Step 1 Move the smaller x-term: subtract $2x$ from both sides $4x+4=20$

Step 2 Subtract 4 from both sides $4x=16$

Step 3 Divide both sides by 4 $\boldsymbol{x=4}$

Check $6(4)+4=28$ and $2(4)+20=28$ ✓

Tip: subtracting the *smaller* x-term keeps the coefficient of x positive.

9. $5x+2=3x+10$

$x=$ _____

10. $7x-3=4x+12$

$x=$ _____

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11. $6x + 1 = 2x + 13$

$x =$ _____

12. $9x - 4 = 5x + 20$

$x =$ _____

13. $4x + 7 = x + 16$

$x =$ _____

14. $8x = 3x + 35$

$x =$ _____

15. $2x + 15 = 5x$

$x =$ _____

16. $10x - 6 = 7x + 9$

$x =$ _____

Type 3 Distribute, Then Variables on Both Sides

Watch Me — Worked Example

Solve $2(x + 5) = x + 13$**.**

Step 1	Distribute	$2x + 10 = x + 13$
Step 2	Subtract x from both sides	$x + 10 = 13$
Step 3	Subtract 10 from both sides	$\boldsymbol{x = 3}$
Check		$2(3 + 5) = 16$ and $3 + 13 = 16$ ✓

17. $2(x + 4) = x + 11$

$x =$ _____

18. $3(x - 1) = x + 7$

$x =$ _____

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19. $4(x+2) = 2x + 14$

$x =$ _____

20. $5(x-2) = 3x + 4$

$x =$ _____

21. $2(x+6) = 4x + 2$

$x =$ _____

22. $3(x+3) = 5x + 1$

$x =$ _____

23. $6(x-1) = 2x + 10$

$x =$ _____

24. $4(x-3) = x + 3$

$x =$ _____

Type 4 Distribute, Then Combine Like Terms

Watch Me — Worked Example

Solve $2(x+4) + 3x = 33$**.**

Step 1 Distribute	$2x + 8 + 3x = 33$
Step 2 Combine like terms ($2x + 3x = 5x$)	$5x + 8 = 33$
Step 3 Subtract 8, then divide by 5	$5x = 25,\ \boldsymbol{x = 5}$
Check	$2(5+4) + 3(5) = 18 + 15 = 33$ ✓

25. $2(x+3) + x = 21$

$x =$ _____

26. $3(x-2) + 2x = 14$

$x =$ _____

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27. $4(x+1)+x=29$

$x=$ _____

28. $2(x-4)+3x=17$

$x=$ _____

29. $5(x+2)-2x=25$

$x=$ _____

30. $3(x+5)+x=31$

$x=$ _____

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